THE WEIL PAIRING, ENDOMORPHISMS AND AUTOMORPHISMS

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Section 1 shows that the determinant of the mod N Galois representation is the cyclotomic character. Section 2 exhibits an endomorphism ring that is an order in a definite quaternion algebra, the third case of the classification [Sil09, Theorem III.9.3], on a single curve over $\mathbb{F}_9$. Section 3 computes the automorphism groups of order 12 and 24 in characteristic 3 and 2, the two cases of [Sil09, Theorem III.10.1] in which the group is not cyclic.

Reading. [Sil09, §§III.8–III.10], with [Sil09, Appendix A.1] for the automorphism groups in characteristic 2 and 3. For the quaternion background: [Voi21, Definition 2.2.1] for the algebra $(a, b \mid \mathbb{Q})$, [Voi21, §§3.2–3.3] for the standard involution, the reduced trace and the reduced norm, [Voi21, Main Theorem 5.4.4 and §14.1] for split and definite algebras, and [Voi21, Definitions 10.2.1 and 10.4.1] for orders and maximal orders. Proposition 2.6, Theorem 2.7 and Proposition 2.8 are the results we actually use.

`Magma`**.** If you have no local installation, every block below runs in the free online calculator, which caps how long a single job may take.

Standing notation.

- A field k, with an algebraic closure $\overline{k}$ and the separable closure $k^{\mathrm{sep}} \subseteq \overline{k}$.
- Multiplication by m, written $[m]$, and the dual $\hat{\varphi}$ of an isogeny φ.
- The q-power Frobenius endomorphism $\mathsf{F} \in \mathrm{End}(E)$ of a curve over $\mathbb{F}_q$. Beware that φ_ℓ already means the map induced by φ on $T_\ell(E)$, so the Frobenius *endomorphism* is F and never φ_q.

How to use this. Allow about 20 minutes for §1, 25 for §2 and 25 for §3. In Problem 1.5 the heart is part (3), and parts (4) and (5) are corollaries to finish at home. In Problem 2.9, parts (1)–(3) are the session and parts (4) and (5) the payoff. They can be started there and finished at home. In Problem 3.5, do characteristic 3 (parts (1)–(3)) in the session and characteristic 2 (parts (4) and (5)) at home: it is the same technique with more bookkeeping. The `Magma` parts need no time in the session: do the hand work there and run the code afterwards, at a keyboard, as a check on what you wrote. `Magma` can be slow to start, and a few of the calls below take a minute or two to return. None of them takes longer.

1. The Weil pairing and the cyclotomic character

Fix $N \geqslant 2$ with $\mathrm{char}\, k \nmid N$, so that $E[N](k^{\mathrm{sep}}) \cong (\mathbb{Z}/N\mathbb{Z})^2$.

Definition 1.1 (The Weil pairing)**.** We write

$$e_N \colon E[N](k^{\mathrm{sep}}) \times E[N](k^{\mathrm{sep}}) \to \mu_N(k^{\mathrm{sep}})$$

for the Weil pairing.

Definition 1.2 (Galois representations)**.** We write

$$\bar{\rho}_{E,N} \colon \mathrm{Gal}_k \to \mathrm{Aut}\big(E[N](k^{\mathrm{sep}})\big) \cong \mathrm{GL}_2(\mathbb{Z}/N\mathbb{Z})$$

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for the mod N Galois representation, the isomorphism depending on a choice of basis, and $\rho_{E,\ell}$ for the ℓ-adic representation on $T_\ell(E)$.

Proposition 1.3 (Properties of the Weil pairing [Sil09, Propositions III.8.1 and III.8.2])**.** *The Weil pairing e_N is*

(1) bilinear;
(2) alternating: $e_N(T,T)=1$, so $e_N(S,T)=e_N(T,S)^{-1}$;
(3) non-degenerate: if $e_N(S,T)=1$ for all $S\in E[N](k^{\mathrm{sep}})$, then $T=O$;
(4) Galois equivariant: $\sigma\big(e_N(S,T)\big)=e_N(\sigma(S),\sigma(T))$ for $\sigma\in\mathrm{Gal}_k$;
(5) compatible: $e_{NN'}(S,T)=e_N([N']S,T)$;
(6) adjoint for isogenies: $e_N(\varphi S,T)=e_N(S,\hat{\varphi}T)$.

These properties are.

Proposition 1.4 (The determinant of φ_ℓ [Sil09, Proposition III.8.6])**.** *Let $\varphi\in\mathrm{End}(E)$ and let $\ell\neq\mathrm{char}\,k$ be a prime. Then $\det\varphi_\ell=\deg\varphi$.*

Problem 1.5 (The determinant is the cyclotomic character)**.**
(1) Let $\zeta\in\mu_N(k^{\mathrm{sep}})$ be primitive. Show that for each $\sigma\in\mathrm{Gal}_k$ there is a unique $\chi_N(\sigma)\in\mathbb{Z}/N\mathbb{Z}$ with $\sigma(\zeta)=\zeta^{\chi_N(\sigma)}$, that $\chi_N(\sigma)$ is a unit, and that the resulting map $\chi_N\colon\mathrm{Gal}_k\to(\mathbb{Z}/N\mathbb{Z})^\times$ is a homomorphism independent of the choice of ζ. This is the mod N cyclotomic character.
(2) Let S,T be a basis of $E[N](k^{\mathrm{sep}})$ and put $\zeta:=e_N(S,T)$. Show that ζ is a *primitive* N-th root of unity, using only parts (1)–(3) of Proposition 1.3.
(3) Write $\sigma(S)=aS+cT$ and $\sigma(T)=bS+dT$, so that $\bar\rho_{E,N}(\sigma)=\left[\begin{smallmatrix}a&b\\c&d\end{smallmatrix}\right]$ in the basis S,T. Expand $e_N(\sigma(S),\sigma(T))$ and deduce

$$\det\bar\rho_{E,N}=\chi_N.$$

(4) Take $k=\mathbb{Q}$. Show that $\det\bar\rho_{E,N}$ is surjective for *every* elliptic curve E over $\mathbb{Q}$, that $\mathbb{Q}(\zeta_N)\subseteq\mathbb{Q}\big(E[N]\big)$, and that $\det\bar\rho_{E,N}(c)=-1$ for c a complex conjugation: the representation is odd.
(5) Take $k=\mathbb{F}_q$ with q prime to N. Show that $\det\bar\rho_{E,N}(\mathsf{F})=q$, and check this against the identity $\det\varphi_\ell=\deg\varphi$.
(6) Check (5) numerically in `Magma`, taking $N=3$, $q=5$ and $E\colon y^2=x^3+1$ over $\mathbb{F}_5$.

```
q := 5;  N := 3;
E := EllipticCurve([GF(q) | 0, 1]);      // y^2 = x^3 + 1
#E;                                      // 6
Ek := ChangeRing(E, GF(q^2));
#Ek, AbelianGroup(Ek);                   // 36, Z/6 + Z/6
T3 := [P : P in Points(Ek) | N*P eq Ek!0];
#T3;                                     // 9
S := [P : P in T3 | P ne Ek!0][1];
T := [P : P in T3 | P ne Ek!0 and P ne S and P ne -S][1];
zeta := WeilPairing(S, T, N);  Order(zeta);    // 3: a primitive cube root
Frob := func<P | P eq Ek!0 select Ek!0 else Ek![P[1]^q, P[2]^q]>;
WeilPairing(Frob(S), Frob(T), N) eq zeta^q;    // true
```

Two lines carry the mathematics. That `Order(zeta)` is 3 is (2). That the last line is `true` is $\det\bar\rho_{E,3}(\mathsf{F})=q$, which is (5) on this curve. Note that `Frob` raises both coordinates to the q-th power: that is the q-power Frobenius, an endomorphism because E is defined over $\mathbb{F}_q$. Now write the matrix of F in the basis S,T and read its determinant off:

```
[ <a, b> : a, b in [0..2] | a*S + b*T eq Frob(S) ];   // [ <1, 0> ]
[ <a, b> : a, b in [0..2] | a*S + b*T eq Frob(T) ];   // [ <0, 2> ]
GF(3)!q;                                             // 2
```

Check that a different choice of basis moves the matrix but not the determinant.

Remark 1.6. Part (3) of Problem 1.5 gives back [Sil09, Corollary III.8.1.1]: if $E[N](k^{\mathrm{sep}}) \subseteq E(k)$, then $\bar{\rho}_{E,N}$ is trivial, so χ_N is trivial, so every $\sigma \in \mathrm{Gal}_k$ fixes a primitive N-th root of unity, and $\mu_N(k^{\mathrm{sep}}) \subseteq k$.

A word, for the curious, on the example in Problem 1.5(6): choosing it was half of that part. We want $q \not\equiv 1 \pmod{N}$, so that χ_N is *not* trivial and the check has content. By Remark 1.6 that already forces $E[N](k^{\mathrm{sep}}) \not\subseteq E(\mathbb{F}_q)$, so the N-torsion becomes rational only over an extension. The curve $E\colon y^2 = x^3 + 1$ over $\mathbb{F}_5$ has $\#E(\mathbb{F}_5) = 6$ and $E(\mathbb{F}_{25}) \cong (\mathbb{Z}/6\mathbb{Z})^2$, so all of $E[3](\overline{\mathbb{F}}_5)$ is already rational over the quadratic extension: small enough to list, large enough to have something to say.

2. A quaternionic endomorphism ring

The classification of [Sil09, Theorem III.9.3] puts the endomorphism ring of an elliptic curve in one of three cases: $\mathbb{Z}$, an order in an imaginary quadratic field, or an order in a definite quaternion algebra over $\mathbb{Q}$. The first two are easy to exhibit: a generic curve has endomorphism ring $\mathbb{Z}$, and $y^2 = x^3 + x$ has $\mathbb{Z}[i]$ inside its endomorphism ring. Here is an example of the third, on the smallest curve that has one.

Definition 2.1 (The quaternion algebra D [Voi21, Definition 2.2.1])**.** Let $D := (-1, -3 \mid \mathbb{Q})$, the quaternion algebra with $i^2 = -1$, $j^2 = -3$ and $ij = -ji$. Its reduced discriminant is the product of the primes p at which $D \otimes_{\mathbb{Q}} \mathbb{Q}_p$ is not split.

Definition 2.2 (Definite quaternion algebras [Voi21, Main Theorem 5.4.4 and §14.1])**.** A quaternion algebra B over $\mathbb{Q}$ is definite when $B \otimes_{\mathbb{Q}} \mathbb{R}$ is Hamilton's $\mathbb{H}$ rather than $M_2(\mathbb{R})$, equivalently when its reduced norm is a positive definite quadratic form in the four coordinates. Otherwise B is indefinite.

Definition 2.3 (Orders and the reduced discriminant [Voi21, §15.2])**.** Orders are as in [Voi21, Definition 10.2.1]. The reduced discriminant $\mathrm{discrd}(\mathcal{O})$ of an order $\mathcal{O} \subseteq D$ is the positive integer with $\mathrm{discrd}(\mathcal{O})^2 = \left|\det\left(\mathrm{trd}(\alpha_k \alpha_l)\right)_{k,l}\right|$ for a $\mathbb{Z}$-basis $\alpha_1, \ldots, \alpha_4$ of $\mathcal{O}$.

Proposition 2.4 (The characteristic polynomial of Frobenius [Sil09, §V.2])**.** *Let E be an elliptic curve over $\mathbb{F}_q$ with F its q-power Frobenius, $a := q + 1 - \#E(\mathbb{F}_q)$, and $\ell \nmid q$ a prime. Then F satisfies $\mathsf{F}^2 - [a]\mathsf{F} + [q] = 0$ in $\mathrm{End}(E)$: by [Sil09, Proposition III.8.6] the characteristic polynomial of F_ℓ is $T^2 - aT + q$, and every $\varphi \in \mathrm{End}(E)$ satisfies $\varphi^2 - [\mathrm{tr}\,\varphi_\ell]\varphi + [\deg \varphi] = 0$ in $\mathrm{End}(E)$, not merely on $T_\ell(E)$.*

Corollary 2.5 (Factoring through a separable isogeny [Sil09, Corollary III.4.11])**.** *If $\varphi\colon E_1 \to E_2$ is separable and $\psi\colon E_1 \to E_3$ is an isogeny with $\ker \varphi \subseteq \ker \psi$, then $\psi = \lambda \circ \varphi$ for a unique isogeny $\lambda\colon E_2 \to E_3$.*

Proposition 2.6 (Where D is split [Voi21, Exercise 11.11 and §14.1])**.** *The algebra $D \otimes_{\mathbb{Q}} \mathbb{Q}_p$ is split for every prime $p \neq 3$, and $D \otimes_{\mathbb{Q}} \mathbb{Q}_3$ is not. So D is a division algebra, and its reduced discriminant is 3.*

Theorem 2.7 (Maximal orders [Voi21, Theorem 15.5.5])**.** *An order $\mathcal{O} \subseteq D$ is maximal if and only if $\mathrm{discrd}(\mathcal{O}) = 3$.*

Proposition 2.8 (The index and the discriminant [Voi21, §15.2]). *If $\mathcal{O} \subseteq \mathcal{O}'$ are orders of D, then* $\operatorname{discrd}(\mathcal{O}) = [\mathcal{O}' : \mathcal{O}] \operatorname{discrd}(\mathcal{O}')$.

For the rest of this section

$$E\colon y^2 = x^3 - x \quad \text{over } \mathbb{F}_3,$$

so that the generic E of Proposition 2.4 is this curve, with $q = 3$.

Problem 2.9 ($\operatorname{End}(E_{\mathbb{F}_9})$ is a maximal order in a quaternion algebra).
Every part below has a hand computation and a check to run in `Magma`. The curve is small enough that $[i]$ and F can be verified on all sixteen points of $E(\mathbb{F}_9)$.
(1) Check that E is an elliptic curve over $\mathbb{F}_3$, list $E(\mathbb{F}_3)$, and deduce that $a = 0$ and that

$$\mathsf{F}^2 = [-3].$$

Record also that every point of $E[2](\overline{\mathbb{F}}_3)$ lies in $E(\mathbb{F}_3)$: this is what makes part (5) work.

```
E3 := EllipticCurve([GF(3) | -1, 0]);    // prints as y^2 = x^3 + 2*x
Points(E3);
#E3;                                     // 4
3 + 1 - #E3;                             // 0        (so a = 0)
jInvariant(E3);                          // 0
```

`Magma` hands you the point count and the j-invariant. The identity $\mathsf{F}^2 = [-3]$ is yours to deduce, from Proposition 2.4. Read $E[2](\overline{\mathbb{F}}_3) \subseteq E(\mathbb{F}_3)$ off the list of rational points — which points of $y^2 = f(x)$ have order 2?
(2) Show that $x^2 + 1$ is irreducible over $\mathbb{F}_3$, so that $\mathbb{F}_9 = \mathbb{F}_3(i)$ with $i^2 = -1$. Show that

$$[i](x, y) \coloneqq (-x, iy)$$

defines an endomorphism of $E_{\mathbb{F}_9}$, and that $[i]^2 = [-1]$. Why is a morphism $E \to E$ fixing O automatically a group homomorphism? Mind Warning 2.10 before you start typing.

```
F9 := GF(9);  i := Sqrt(F9!(-1));
i^2, MinimalPolynomial(i);               // 2 ($.1^2 + 1), and 2 = -1
E9 := ChangeRing(E3, F9);  #E9;          // 16
imap := func<P | P eq E9!0 select E9!0 else E9![-P[1], i*P[2]]>;
forall{P : P in Points(E9) | imap(P) in Points(E9)};        // [i] is well defined
forall{<P,Q> : P, Q in Points(E9) | imap(P+Q) eq imap(P)+imap(Q)};  // additive
forall{P : P in Points(E9) | imap(imap(P)) eq -P};          // [i]^2 = [-1]
// all true
```

The second `forall` is additivity, checked on all 16^2 pairs. That is a proof only because the curve is finite, and it is no answer to the question above: additivity is not a coincidence to be confirmed point by point but a consequence of fixing O.
(3) Show that $\mathsf{F}[i] = -[i]\mathsf{F}$. Conclude that $\operatorname{End}(E_{\mathbb{F}_9})$ is not commutative, so that $E_{\mathbb{F}_9}$ falls into the third case of the classification.

```
Frob := func<P | P eq E9!0 select E9!0 else E9![P[1]^3, P[2]^3]>;
forall{P : P in Points(E9) | Frob(Frob(P)) eq -3*P};             // Frob^2 = [-3]
forall{P : P in Points(E9) | Frob(imap(P)) eq -imap(Frob(P))};   // Frob[i] = -[i]Frob
// both true
```

The first line verifies the $\mathsf{F}^2 = [-3]$ you deduced in (1) from the characteristic polynomial, which is worth having independently. For the second, do the computation by hand as well: it comes down to a single fact about i^3.

(4) Deduce that

$$\mathrm{End}(E_{\mathbb{F}_9}) \otimes_{\mathbb{Z}} \mathbb{Q} \cong D = (-1, -3 \mid \mathbb{Q}).$$

Write down the reduced norm of D and check that D is definite. How does the reduced norm compare with the degree of an endomorphism?

```
D<ii,jj> := QuaternionAlgebra<Rationals() | -1, -3>;
RamifiedPrimes(D), Discriminant(D);      // [ 3 ] 3
IsDefinite(D);                           // true
Trace(jj), Norm(jj);                     // 0 3
```

The two arguments are the squares of the generators, so the algebra `Magma` builds is the D of Definition 2.1. Here `Trace` and `Norm` on a quaternion algebra are the reduced trace and the reduced norm, and the pair they return for `jj` should be recognizable as $(\mathrm{tr}\,\mathsf{F}, \deg \mathsf{F})$. All of this is about D alone. That D is the *whole* of $\mathrm{End}(E_{\mathbb{F}_9}) \otimes \mathbb{Q}$ is the one claim here `Magma` cannot be asked, and it needs a rank count.

(5) Show that $\frac{1+\mathsf{F}}{2}$ and $\frac{i+i\mathsf{F}}{2}$ lie in $\mathrm{End}(E_{\mathbb{F}_9})$.[a] Then show that

$$\mathcal{O} := \mathbb{Z} + \mathbb{Z}i + \mathbb{Z}\frac{1+\mathsf{F}}{2} + \mathbb{Z}\frac{i+i\mathsf{F}}{2}$$

is an order of D, compute $\mathrm{discrd}(\mathcal{O})$, and conclude with Theorem 2.7 that $\mathrm{End}(E_{\mathbb{F}_9}) = \mathcal{O}$ is a maximal order.

```
OO := QuaternionOrder([D!1, ii, jj, ii*jj]);
Discriminant(OO);                        // 12
O  := QuaternionOrder([D!1, ii, (1+jj)/2, (ii+ii*jj)/2]);
Discriminant(O), IsMaximal(O);           // 3 true
```

`QuaternionOrder` takes a $\mathbb{Z}$-basis and `Discriminant` of an order is its reduced discriminant, so the two lists above are the $\mathbb{Z}$-bases of $\mathcal{O}_0$ and $\mathcal{O}$. Do not reach for `Index` to compare the two orders: `Magma` has no index for quaternion orders, and answers `Runtime error in 'Index': Bad argument types`. Recover $[\mathcal{O} : \mathcal{O}_0]$ from the two discriminants instead, by Proposition 2.8, and check it against the determinant of the change of basis.

The first sentence of this part is where the real work is, and none of the code touches it: $\frac{1+\mathsf{F}}{2}$ is a quotient in a ring with no division, and only the kernel argument puts it in $\mathrm{End}(E_{\mathbb{F}_9})$.

[a] *Hint:* use Corollary 2.5 with $\varphi = [2]$.

Warning 2.10 ($\mathbb{F}_9$ does not come with i)**.** `GF(9)` does *not* come with i. `Magma`'s default generator of $\mathbb{F}_9$ is a root of the Conway polynomial $x^2 + 2x + 2$, so writing `F9<i> := GF(9)` gives an i with $i^2 = i + 1$, not $i^2 = -1$, and every check in Problem 2.9 would silently fail. Construct the square root of -1 explicitly, and confirm it: `MinimalPolynomial` of the element you build must be $x^2 + 1$, and `i^2` must print as 2, which is -1 in $\mathbb{F}_9$.

3. Automorphisms in characteristic 2 and 3

For $\operatorname{char} k = 2, 3$ the geometric automorphism group of an elliptic curve may be non-abelian, which is the subject of Problem 3.5. In the following statements, E is an elliptic curve over k, given by a Weierstrass equation with coefficients a_1, a_2, a_3, a_4, a_6.

Definition 3.1 (Changes of variables)**.** A change of variables preserving a Weierstrass equation is a substitution

$$x = u^2x' + r, \qquad y = u^3y' + u^2sx' + t, \qquad u \in \overline{k}^\times,\ r, s, t \in \overline{k}. \tag{3.1}$$

For such u, r, s, t we write

$$\Psi_{u,r,s,t}(x, y) \coloneqq (u^2x + r,\ u^3y + u^2sx + t).$$

Theorem 3.2 (The automorphism group [Sil09, Theorem III.10.1])**.** *The group* $\operatorname{Aut}(E_{\overline{k}})$ *is finite, of order*

$$\#\operatorname{Aut}(E_{\overline{k}}) = \begin{cases} 2 & \text{if } j(E) \neq 0, 1728, \\ 4 & \text{if } j(E) = 1728 \text{ and } \operatorname{char} k \neq 2, 3, \\ 6 & \text{if } j(E) = 0 \text{ and } \operatorname{char} k \neq 2, 3, \\ 12 & \text{if } j(E) = 0 = 1728 \text{ and } \operatorname{char} k = 3, \\ 24 & \text{if } j(E) = 0 = 1728 \text{ and } \operatorname{char} k = 2. \end{cases}$$

Corollary 3.3 (The cyclic cases [Sil09, Corollary III.10.2])**.** *If* $\operatorname{char} k \neq 2, 3$*, then* $\operatorname{Aut}(E_{k^{\mathrm{sep}}}) \cong \mu_n(k^{\mathrm{sep}})$ *with* $n = \#\operatorname{Aut}(E_{\overline{k}})$*, via* $[\zeta](x, y) \coloneqq (\zeta^2x, \zeta^3y)$*. In particular, the geometric automorphism group is cyclic in that case.*

Proposition 3.4 (The action on the coefficients [Sil09, Proposition III.3.1(b) and Table 3.1])**.** *Under a change of variables* (3.1) *the coefficients transform by*

$$\begin{aligned} ua_1' &= a_1 + 2s, \\ u^2a_2' &= a_2 - sa_1 + 3r - s^2, \\ u^3a_3' &= a_3 + ra_1 + 2t, \\ u^4a_4' &= a_4 - sa_3 + 2ra_2 - (t + rs)a_1 + 3r^2 - 2st, \\ u^6a_6' &= a_6 + ra_4 + r^2a_2 + r^3 - ta_3 - t^2 - rta_1. \end{aligned} \tag{3.2}$$

A substitution (3.1) *carries E to the curve with coefficients a_i', so it is a change of coordinates from E to itself exactly when $a_i' = a_i$ for every i, and then $\Psi_{u,r,s,t}$ is the automorphism of E that it defines. Setting $a_i' = a_i$ in* (3.2) *therefore turns those five lines into five equations in the four unknowns u, r, s, t, whose solutions are exactly the elements of* $\operatorname{Aut}(E_{\overline{k}})$*.*

The whole point of this section is visible in (3.2): observe the 2's and 3's in the coefficients. The characteristic enters through the terms $2s$, $2t$ and $3r$. Away from 2 and 3 they are invertible multiples of s, t and r, which determines those three and leaves u as the only free parameter: that is why the group is cyclic there. In characteristic 3 the term $3r$ vanishes, in characteristic 2 the terms $2s$ and $2t$ do, and the parameters they no longer pin down are what make the two groups of Problem 3.5 large, and non-cyclic.

Problem 3.5 (The two non-cyclic cases)**.**
As in Problem 2.9, each part pairs a hand computation with a check in `Magma`. Here the machine is best at naming the group, which is the trickiest thing to do by hand.

(1) Let $\operatorname{char} k = 3$. Show that every short Weierstrass equation $y^2 = x^3 + Ax + B$ has $j = 0$, and explain why a curve with $j \neq 0$ in characteristic 3 needs an a_2x^2 term. So the case "$j = 0 = 1728$, $\operatorname{char} k = 3$" of Theorem 3.2 is exactly the short-Weierstrass one. The first claim is a one-line sweep over the whole family:

```
{ jInvariant(EllipticCurve([GF(3) | A, B])) : A, B in GF(3) | A ne 0 };  // { 0 }
jInvariant(EllipticCurve([GF(3) | 0, 1, 0, 0, -1]));   // 1
```

The condition `A ne 0` is what keeps the equation non-singular: work out what $\Delta \neq 0$ becomes for $y^2 = x^3 + Ax + B$ once $27 = 0$, and you will find that B drops out. The set that comes back has one element, which is the first claim. The second line is a curve with an a_2x^2 term, and its j-invariant is not in that set.

(2) Let $\operatorname{char} k = 3$ and $E\colon y^2 = x^3 - x$. Using (3.2), show first that $s = t = 0$, and then that (3.1) is an automorphism of E if and only if

$$u^4 = 1 \qquad \text{and} \qquad r^3 - r = 0,$$

i.e. $u \in \mu_4(\overline{\mathbb{F}}_3) \subseteq \mathbb{F}_9^\times$ and $r \in \mathbb{F}_3$. Conclude $\#\operatorname{Aut}(E_{\overline{\mathbb{F}}_3}) = 12$, and that all 12 are already defined over $\mathbb{F}_9$. Where did r become free, and what would have forced $r = 0$ if $\operatorname{char} k \neq 3$? Two counts settle both claims at once:

```
E3 := EllipticCurve([GF(3) | -1, 0]);
#AutomorphismGroup(E3), #AutomorphismGroup(ChangeRing(E3, GF(9)));   // 6 12
```

Predict both numbers from your conditions on u and r before running it. Over $\mathbb{F}_3$ the fourth roots of unity are only ± 1, while r already ranges over all of $\mathbb{F}_3$. Over $\mathbb{F}_9$ all four roots of unity appear. The second number being 12, and being reached over $\mathbb{F}_9$ rather than over some larger field, is exactly the claim that all twelve automorphisms are defined there.

(3) Write $\varphi_{u,r} := \Psi_{u,r,0,0}$, so that $\varphi_{u,r}(x, y) = (u^2x + r, u^3y)$. Compute $\varphi_{u',r'} \circ \varphi_{u,r}$ and deduce

$$\operatorname{Aut}(E_{\overline{\mathbb{F}}_3}) \cong \mathbb{Z}/3\mathbb{Z} \rtimes \mathbb{Z}/4\mathbb{Z},$$

where $\mathbb{Z}/3\mathbb{Z} = \mathbb{F}_3$ is the group of translations r, where $\mathbb{Z}/4\mathbb{Z} = \mu_4(\overline{\mathbb{F}}_3)$ is the group of the u, and where u acts on r by $r \mapsto u^2 r$. Since $u^4 = 1$ forces $u^2 = \pm 1$, that action is not faithful: it factors through the squaring map $\mu_4(\overline{\mathbb{F}}_3) \twoheadrightarrow \{\pm 1\} = \operatorname{Aut}(\mathbb{Z}/3\mathbb{Z})$, so ± 1 act trivially and $\pm i$ act by $r \mapsto -r$. Show this group is non-abelian with a *unique* element of order 2, and deduce that it is neither cyclic, nor dihedral, nor A_4: it is the dicyclic group of order 12. Which automorphism is the unique involution? `AutomorphismGroup` returns a polycyclic group and `IdentifyGroup` gives its label in the small-groups database.

```
G3 := AutomorphismGroup(ChangeRing(E3, GF(9)));
IdentifyGroup(G3), GroupName(G3);        // <12, 1>  C3:C4
IsAbelian(G3), IsCyclic(G3);             // false false
{* Order(g) : g in G3 *};                // {* 1, 2, 3^^2, 4^^6, 6^^2 *}
```

The last line is the multiset of element orders. Say how it rules out the dihedral group and A_4 — how many involutions does each of those have? — and match its single element of order 2 against the automorphism you named.

(4) Let $\operatorname{char} k = 2$ and $E\colon y^2 + y = x^3$. Using (3.2), show that $a_1' = a_1$ automatically, that $a_2' = 0$ forces $r = s^2$, and then that (3.1) is an automorphism if and only if

$$u^3 = 1, \qquad s^4 = s, \qquad t^2 + t = s^6.$$

Deduce that $s \in \mathbb{F}_4$, count the solutions, and conclude $\#\operatorname{Aut}(E_{\overline{\mathbb{F}}_2}) = 24$, all defined over $\mathbb{F}_4$. The same two counts as in (2):

```
E2 := EllipticCurve([GF(2) | 0, 0, 1, 0, 0]);          // y^2 + y = x^3
#AutomorphismGroup(E2), #AutomorphismGroup(ChangeRing(E2, GF(4)));   // 2 24
```

Predict both first. Over $\mathbb{F}_2$ the three conditions bite hard: $u^3 = 1$ leaves only $u = 1$, and of the values $s \in \mathbb{F}_2$ one of them makes $t^2 + t = s^6$ unsolvable in $\mathbb{F}_2$. The gap between the two numbers here is much wider than in characteristic 3, and that gap is the content of "all defined over $\mathbb{F}_4$".

(5) Write $\psi_{s,t} \coloneqq \Psi_{1,s^2,s,t}$ for the automorphisms with $u = 1$, so that $\psi_{s,t}(x, y) = (x + s^2, y + sx + t)$. Show that

$$\psi_{s',t'} \circ \psi_{s,t} = \psi_{s+s',\, t+t'+s's^2}, \qquad \psi_{s,t}^2 = \psi_{0,s^3},$$

and deduce that these 8 automorphisms form a copy of the quaternion group Q_8. For the remaining factor, let $\omega \in \mathbb{F}_4$ be a primitive cube root of unity and $\varphi_\omega \coloneqq \Psi_{\omega,0,0,0}$, which has order 3. Check that conjugation by it is

$$\varphi_\omega \circ \psi_{s,t} \circ \varphi_\omega^{-1} = \psi_{\omega s,\, t},$$

so φ_ω normalizes Q_8, fixes its center $\langle \psi_{0,1} \rangle$, and cycles the three pairs $\pm i, \pm j, \pm k$ — the order-3 automorphism of Q_8. Since $\langle \varphi_\omega \rangle \cap Q_8 = 1$, conclude that

$$\operatorname{Aut}(E_{\overline{\mathbb{F}}_2}) \cong Q_8 \rtimes \mathbb{Z}/3\mathbb{Z} \cong \mathrm{SL}_2(\mathbb{F}_3).$$

As before:

```
G2 := AutomorphismGroup(ChangeRing(E2, GF(4)));
IdentifyGroup(G2), GroupName(G2);        // <24, 3>  SL(2,3)
{* Order(g) : g in G2 *};                 // {* 1, 2, 3^^8, 4^^6, 6^^8 *}
```

Exactly one element of order 2 again. Explain why that alone pins down Q_8 among the groups of order 8, and why it does *not* by itself pin down $\mathrm{SL}_2(\mathbb{F}_3)$ among the groups of order 24 — what extra fact did you use?

Remark 3.6 (Both groups are unit groups)**.** Automorphisms are the units of the endomorphism ring, $\operatorname{Aut}(E_{\overline{k}}) = \operatorname{End}(E_{\overline{k}})^\times$. In characteristic 2 and 3 the curve with $j = 0$ has $\operatorname{End}(E_{\overline{k}})$ a maximal order in the quaternion algebra over $\mathbb{Q}$ that is not split exactly at p and ∞ [Sil09, Theorem V.3.1], [Voi21, Theorem 42.1.9]. For $p = 3$ that order is the one computed in Problem 2.9(5), and its unit group is the dicyclic group of order 12 of Problem 3.5(3). For $p = 2$ it is the Hurwitz order $\mathbb{Z} + \mathbb{Z}i + \mathbb{Z}j + \mathbb{Z}\frac{1+i+j+ij}{2}$ in $(-1, -1 \mid \mathbb{Q})$ [Voi21, §11.1], whose unit group is the binary tetrahedral group of order 24 of Problem 3.5(5). This explains the shape of Theorem 3.2: when $\operatorname{End}(E_{\overline{k}})$ is commutative it is $\mathbb{Z}$ or an imaginary quadratic order, whose unit groups are the cyclic groups of orders 2, 4 and 6, giving its first three lines, while a non-commutative $\operatorname{End}(E_{\overline{k}})$ is a quaternion order, and its unit group need not be, and here is not, cyclic.

Both counts can be recovered from the orders alone, with no elliptic curve anywhere in the computation:

```
for d in [-1, -3] do
  Dd := QuaternionAlgebra<Rationals() | -1, d>;
  Od := MaximalOrder(Dd);
  print RamifiedPrimes(Dd), #UnitGroup(Od);
end for;
// [ 2 ] 24      <- Hurwitz order, characteristic 2
// [ 3 ] 12      <- the endomorphism ring of Problem 2.9, characteristic 3
```

So the 24 and the 12 of Theorem 3.2 are the unit counts of the two maximal quaternion orders.

Looking ahead. Curves whose Frobenius has $a = 0$, like the one of section 2, have a name and a theory of their own. They come later in the course, with Hasse's bound and the structure of End over $\mathbb{F}_q$. In characteristic 0 the third case of the classification never occurs, and $\mathrm{End}(E_{\overline{k}}) \neq \mathbb{Z}$ becomes complex multiplication. The oddness of $\rho_{E,\ell}$ from Problem 1.5(4) is the first hint of modularity.

References

[Sil09] Joseph H. Silverman, *The arithmetic of elliptic curves*, Second, Graduate Texts in Mathematics, vol. 106, Springer, Dordrecht, 2009. MR2514094 ↑

[Voi21] John Voight, *Quaternion algebras*, Graduate Texts in Mathematics, vol. 288, Springer, Cham, 2021. MR4279905 ↑